

Mehul Sharma

Full-stack software engineer

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PROFILE

AI Engineer and backend-focused software engineer with experience building **production-grade LLM systems**, including **RAG pipelines, agent-based architectures, and personalized AI assistants**. Strong background in **Python, FastAPI, PostgreSQL, and ML systems**, with hands-on experience in **LangChain, model evaluation, and observability**. Proven ability to design scalable, secure backend services and deploy AI solutions used in real enterprise environments.

EDUCATION

Computing & Software Systems

2022 – 2025

University of Melbourne, Australia

- University College 2022-2023
- Cricket Team, Football Team

International Baccalaureate

2019 – 2021

National Public School, Singapore

- Football Team, Quiz Team

WORK EXPERIENCE

AI Engineer

12/2025 – Present

Drish Infotech

- Built an **internal employee AI assistant** using **RAG, LangChain, and agent-based architectures** to deliver personalized, role-aware responses across organizational knowledge.
- Integrated **Model Context Protocol (MCP)** to inject secure user context (role, department, permissions), enabling **controlled personalization** at scale.
- Designed document ingestion and retrieval pipelines for internal policies and SOPs, improving response accuracy and reducing manual support effort.

Co-Founder / Software Engineer

02/2025 – 09/2025

Stealth AI Startup

Melbourne, Australia

- Built an end-to-end AI-powered platform enabling users to generate custom T-shirt designs from natural language prompts using text-to-image diffusion models.
- Designed and implemented scalable backend services in Python and FastAPI for prompt processing, image generation orchestration, and asset storage.
- Led rapid product experimentation end-to-end, validating user demand and technical feasibility through real customer usage before winding down the venture.

Backend Engineer

06/2024 – 01/2025

Curtin Health

Melbourne, Australia

- Built a **scalable backend system** supporting nurses, doctors, and patients across multiple facilities, handling **500+ daily health test requests**.
- Integrated **15+ clinical test workflows** with standardized data models and a **normalized PostgreSQL schema (20+ tables)**, reducing data redundancy by **40%**.
- Developed **secure REST APIs** with JWT-based authentication and role-based access control, ensuring **HIPAA-compliant data handling**.

PROJECT

AI Resume Sorter

(Python, TensorFlow, FastAPI, PostgreSQL)

- Built an **AI-powered resume-job matching microservice** achieving **85% accuracy**.
- Applied **NLP techniques** (TF-IDF and cosine similarity) to score candidate-job fit, processing **100+ resumes in under 2 seconds**.

- Exposed **REST APIs** with Swagger/OpenAPI documentation and **Dockerized** the service for consistent deployment.
- Designed a **PostgreSQL schema** to store parsed resume data and matching history.

Multi-Tenant RAG Platform (Backend, FastAPI)

(Python, FastAPI, PostgreSQL (pgvector), Redis, LangChain, Docker, Prometheus)

- Built a **production-grade multi-tenant RAG platform** with async ingestion and hybrid retrieval (BM25 + vector + cross-encoder).
- Implemented **citation-enforced, streaming LLM responses** with strict context budgeting.
- Developed an **evaluation framework** for faithfulness, relevance, and completeness with regression tracking.
- Added **production observability** with structured logging, tracing, and Prometheus metrics.

End-to-End ML Platform with Monitoring & Drift Detection

(Python, FastAPI, scikit-learn/XGBoost, Sentence Transformers, MLflow, PostgreSQL, Docker, Prometheus)

- Built an **end-to-end ML platform** for resume–job matching covering training, versioning, deployment, monitoring, and controlled redeployments.
- Designed **feature pipelines** combining structured signals with text embeddings and evaluated models using **ROC-AUC and Precision@K**.
- Implemented **MLflow-based experiment tracking and model versioning** for reproducible training and safe production promotion.
- Deployed a **FastAPI inference service** with schema validation, versioned model loading, and **shadow deployment** support.
- Added **data and prediction drift detection**, exposing metrics to Prometheus and enabling feedback-driven retraining.

SKILLS

Languages

Python (Advanced) JavaScript (Advanced) Java (Intermediate) TypeScript SQL

Frameworks & Libraries

React Node.js FastAPI Next.js Express.js Tailwind CSS TensorFlow

Databases

PostgreSQL SQLite MongoDB

Tools & Platforms

Docker Git GitHub Actions AWS (EC2, S3) Vercel Postman Swagger

Core Competencies

REST API Development JWT Authentication Database Design CI/CD Agile/Scrum Unit Testing Design Patterns (GRASP, GoF) NLP Microservices Architecture